Lisa Alazraki

lisa.alazraki20@imperial.ac.uk • Website • Scholar • GitHub • Linkedin • EU Citizen, UK Settled Status

Final-year PhD student in machine learning and NLP at Imperial College London. My research aims to build AI systems that reason robustly and generalise across tasks, with a focus on agents and multi-agent systems and AI for accelerating scientific research.

EDUCATION

Imperial College London, *PhD Computing* — Supervisor: Marek Rei — *2022-2027*
Imperial College London, *MSc Computing (AI and Machine Learning)* — Classification: Distinction — *2020-2021*
The Open University, *Grad. Cert. Theoretical Statistics and Probability* — Classification: Distinction — *2019-2020*
The Open University, *BSc (Hons) Computing & IT and Mathematics* — Classification: 1st Class — *2015-2019*

EXPERIENCE

Ai2, *Research Associate* • Remote — *Sep - Dec 2026*
Research Intern • Seattle, USA — *Jun - Sep 2026*

- Manager: Varsha Kishore. Team: Asta – AI for Science.
- Post-training LLM deep research agents with multi-turn RL for exhaustive, high-recall search.

Meta, *Research Scientist Intern* • London, UK — *Jun - Dec 2025*

- Manager: Akhil Mathur. Team: Post-training Reasoning, Agents.
- Studied how LLM agent capabilities scale as a function of FLOPs and completion time horizon.
- Designed a marketplace-inspired adaptive multi-agent system that coordinates small agents to push the accuracy/cost Pareto frontier.
- First-authored a paper published at ICML 2026.

Cohere, *Research Intern* • London, UK — *Jun - Dec 2024*

- Manager: Max Bartolo. Team: Command Post-training.
- Developed a reinforcement learning pipeline for reverse engineering human preferences that boosts LLM-as-a-judge evaluation.
- Showed that LLMs attain higher accuracy without explicit corrective feedback, through implicit in-context learning.
- First-authored two papers published at NeurIPS 2025 and EMNLP 2025 respectively.

Google, *Research Intern* • Amsterdam, Netherlands — *Jun - Sep 2023*

- Manager: Thomas Mensink. Team: Perception.
- Developed a model-ensembling framework for knowledge-intensive VQA, outperforming SOTA by 5% on Encyclopedic-VQA.
- First-authored a paper presented at ICBINB at NeurIPS 2023, published in PMLR.

Google, *Student Researcher* • London, UK — *Oct - Dec 2022*
Research Intern • Zurich, Switzerland — *Jun - Sep 2022*

- Manager: Hamza Harkous. Team: Applied Privacy Research.
- Built a retrieval-augmented generation pipeline for user issue classification deployed to production, with 10× recall improvement.
- Granted a global patent as co-inventor of the overall system.

SELECTED PAPERS

Rethinking Rubric Generation to Improve LLM Judge and Reward Modeling for Open-ended Tasks, *NeurIPS 2026* — *2026*
W. F. Shen, X. Qiu, C. Whitehouse, L. Alazraki, S. Goel, F. Barbieri, T. Willi, A. Mathur, I. Leontiadis

Repetition Mismatch: Why Data Mixture Experiments Don't Scale and How to Fix Them, *EMNLP 2026* — *2026*
K. Zhou, L. Alazraki, K. Cao, M. Rei

Scaling Small Agents Through Strategy Auctions, *ICML 2026* — *2026*
L. Alazraki, W. F. Shen, Y. Bachrach, A. Mathur

AgentCoMa: A Compositional Benchmark Mixing Commonsense and Mathematical Reasoning, *ACL 2026 (Oral)* — *2025*
L. Alazraki, L. Chen, A. Brassard, J. Stacey, H. A. Rahmani, M. Rei

Reverse Engineering Human Preferences with Reinforcement Learning, *NeurIPS 2025 (Spotlight)* — *2025*
L. Alazraki, T. Yi-Chern, J. A. Campos, M. Mozes, M. Rei, M. Bartolo

No Need for Explanations: LLMs Can Implicitly Learn from Mistakes In-context, *EMNLP 2025 (Oral)* — *2025*
L. Alazraki, M. Mozes, J. A. Campos, T. Yi-Chern, M. Rei, M. Bartolo

Enhancing LLM Robustness to Perturbed Instructions: An Empirical Study, *ICLR 2025 BuildingTrust* — *2025*
A. Agrawal*, L. Alazraki*, S. Honarvar, M. Rei (*Equal contribution)

Meta-reasoning Improves Tool Use in Large Language Models, *NAACL 2025 Findings* — *2024*
L. Alazraki, M. Rei

How (not) to ensemble LVLMs for VQA, *NeurIPS 2023 ICBINB / PMLR 239* — *2023*
L. Alazraki, L. Castrejon, M. Dehghani, F. Huot, J. Uijlings, T. Mensink

AWARDS

Placement Award, *Alan Turing Institute* *2024*
Selected as one of ∼30 resident PhD students for the Institute's Enrichment Scheme 2024/25.

ICC Poster Competition First Prize, *Imperial College London* *2024*
Awarded for best poster presentation at the Imperial Computing Conference.

IET Postgraduate Prize, *Institution for Engineering and Technology* *2024*
Awarded for 'excellence in postgraduate research' at the IET Postgraduate Research Awards.

Best Student Paper Award, *Institute of Electrical and Electronics Engineers* *2023*
Awarded at the IEEE International Conference on Cognitive Machine Intelligence.

Distinguished Dissertation Award, *Imperial College London* *2021*
Awarded for 'outstanding MSc dissertation in terms of technical achievement and presentation'.

Faculty Commendation, *The Open University* *2017*
Awarded for achieving one of the highest results out of ∼500 students in Algorithms, Data Structures and Computability.

Leslie Walshaw Award, *The Open University* *2016*
Awarded for achieving the highest overall grade of all students in the Greater London region in mathematics module MST125.

LEADERSHIP AND COMMUNITY

Supervision	Conceived 10+ MRes/MSc/MEng thesis specifications and co-supervised 16 students across machine learning and NLP
Mentorship	Hold regular office hours advising applicants on graduate admissions, particularly from non-traditional backgrounds
Teaching	Graduate teaching assistant across six courses spanning computer science, machine learning, deep learning, and NLP
Reviewing	Reviewer for AAAI, ACL Rolling Review, COLING, COLM, ICML (Gold Reviewer), NeurIPS, TMLR, as well as various workshops

SELECTED TALKS

Invited

Strategy Auctions for Agent Routing and Self-Improvement, *NLIP Seminar Series, University of Cambridge* *2026*
Reverse Engineering Human Preferences with Reinforcement Learning, *Lancaster AI, Lancaster University* *2026*
Reverse Engineering Human Preferences with Reinforcement Learning, *Breaking Topics in AI Conference, I-X Imperial* *2025*
Benchmarking LLMs on Mixed-type Compositional Reasoning, *Open Science Community Lightning Talks, Cohere Labs* *2025*

Contributed

Scaling Small Agents Through Strategy Auctions, *Imperial NLP, Imperial College London* *2026*
AgentCoMa, *Annual Meeting of the Association for Computational Linguistics* *2026*
Mixing Commonsense and Mathematical Reasoning in Real-World Scenarios, *NLP Special Interest Group, Alan Turing Institute* *2026*
Pattern Matching vs. Procedural Reasoning in Language Models, *Imperial NLP, Imperial College London* *2026*
Plan Auctions for Task-Aware Selection of LLM Agents Across Scales, *Meta Superintelligence Labs Tech Talks, Meta* *2025*
No Need for Explanations: LLMs Can Implicitly Learn from Mistakes In-context, *Conference on Empirical Methods in NLP* *2025*
Overcoming the Limitations of Autoregressive Generation, *Imperial NLP, Imperial College London* *2024*
A Retrieval-Augmented Generation Pipeline for User Issue Classification, *T5 and Beyond Monthly Talks, Google* *2022*

GRANTS AND SCHOLARSHIPS

IX-WAI Early Career Development Grant, *I-X Imperial* *2025*
OpenAI API Researcher Access Grant, *OpenAI* *2025*
Annual Grant, *Sir Richard Stapley Trust* *2024*
General Fund Grant, *Imperial College Trust* *2023*
IET International Travel Award, *Institution for Engineering and Technology* *2023*
Annual Grant, *Sir Richard Stapley Trust* *2023*
EPSRC Doctoral Scholarship, *Imperial College London* *2022*
DeepMind MSc Scholarship, *Imperial College London* *2020*

PATENTS

Deep Learning System for Navigating Feedback, *US Patent US20250272485A1, CN Patent CN119053957A, WO Patent WO2023224672A1*
Inventors: H. Harkous, S. T. Peddinti, R. Khandelwal, A. Srivastava, N. Taft, L. Alazraki. Assignee: Google LLC

TRAINING AND SKILLS

Summer schools attended	Oxford Machine Learning Summer School MLx Representation Learning & GenAI 2025 + MLx Health & Bio 2025
Programming languages	Python, TypeScript, JavaScript, Java, Lua, MATLAB/Octave, Maxima, C++, Solidity, Prolog, Unix/Bash, HTML, CSS
Libraries / frameworks	PyTorch, TensorFlow, Transformers, verl, vLLM, NumPy, Pandas, Scikit-learn, Jinja2, Matplotlib, Plotly, WanDB